



Proxmox: Storage, Backup, Capacity, and Naming

Purpose: Documents the virtualisation host's storage layout, the backup and recovery strategy, the capacity plan for the planned estate, and the lab-wide machine naming convention. Written so that a reviewer can see not just *what* was configured but *why*, and which controls it satisfies.

Host: proxmox-01 (192.168.10.6), Proxmox VE 9.2.2, standalone node.

Status: Backup configured and verified 2026-08-30.

1. Host overview

Resource	Value
CPU	8 cores/threads
RAM	~32 GiB
Storage	2 physical disks, 2.67 TiB total
Cluster	Standalone (no cluster)

RAM is the binding constraint for how many guests run at once; CPU and storage are abundant.

The screenshot displays the Proxmox VE 9.2.2 web interface. The left sidebar shows the 'Datacenter' view with a tree structure including 'proxmox-01' and its associated VMs and storage. The main panel is titled 'Health' and shows a green checkmark indicating the system is healthy. Below this, the 'Nodes' section shows 1 online node and 0 offline nodes. The 'Guests' section shows 0 running and 2 stopped virtual machines, and 0 running and 0 stopped LXC containers. The 'Resources' section shows CPU, Memory, and Storage usage. At the bottom, a 'Tasks' table shows a recent task: 'Update package database' completed successfully at 09:06:44.

Start Time	End Time	Node	User name	Description	Status
Aug 29 09:06:44	Aug 29 09:06:48	proxmox-01	root@pam	Update package database	OK

Figure 14.1: Datacenter summary: a healthy standalone node (`proxmox-01`), no cluster, with the current guests. This is the top-level health and inventory view.

2. Storage layout

Disk	Size	Role
<code>/dev/sda</code>	3 TB	Proxmox OS + primary guest storage (<code>local</code> , <code>local-lvm</code>)
<code>/dev/sdb</code>	500 GB	Backup target (<code>backup</code> , ext4 Directory storage)

`sda` holds the two default install storages: `local` (ISO images, templates, backups directory) and `local-lvm` (VM/CT disk images). `sdb` was previously an OS disk (leftover EFI + ext4 partitions); it was wiped and repurposed as a dedicated backup store.

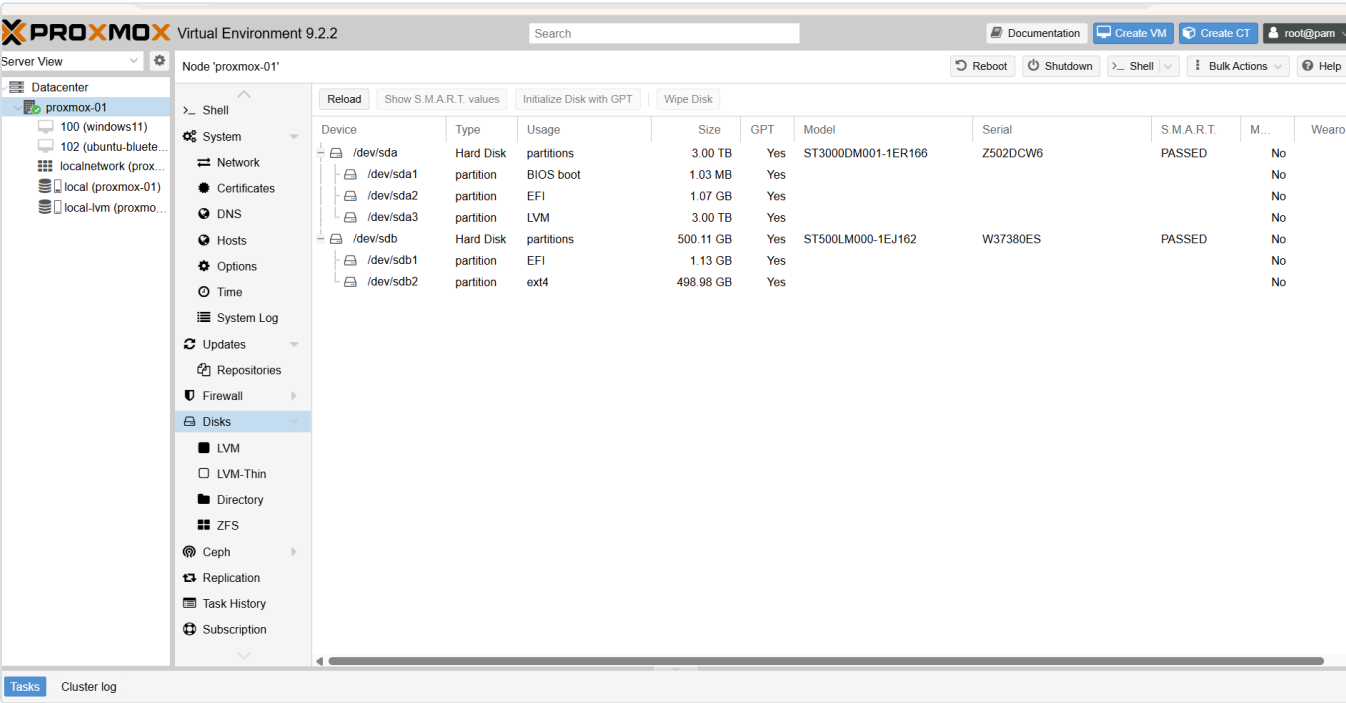


Figure 14.2: The two physical disks: `/dev/sda` (3 TB, Proxmox OS + LVM guest storage) and `/dev/sdb` (500 GB), before `sdb` was repurposed. Both report SMART "PASSED".

Why backups go on a separate disk

Keeping backups on a *different physical disk*

from the running VMs means a failure or corruption of the primary disk does not take the backups with it. This is a basic tenet of recoverable design: the backup must survive the thing it is protecting against.

3. Backup and recovery strategy

The principle

A running VM is not a backup of itself. Snapshots help with quick rollback, but a real backup is an independent, retained copy that survives disk loss, a bad change, or a corrupted guest. Two recent physical-machine failures in this lab (a server NIC, and the Wazuh host losing power) made the point concretely: **anything not backed up is one hardware fault away from a rebuild.**

What was configured (2026-08-30)

1. **Wiped** `/dev/sdb` and created a **Directory storage** named `backup` (ext4, mounted at `/mnt/pve/backup`), content type *VZDump backup file*.

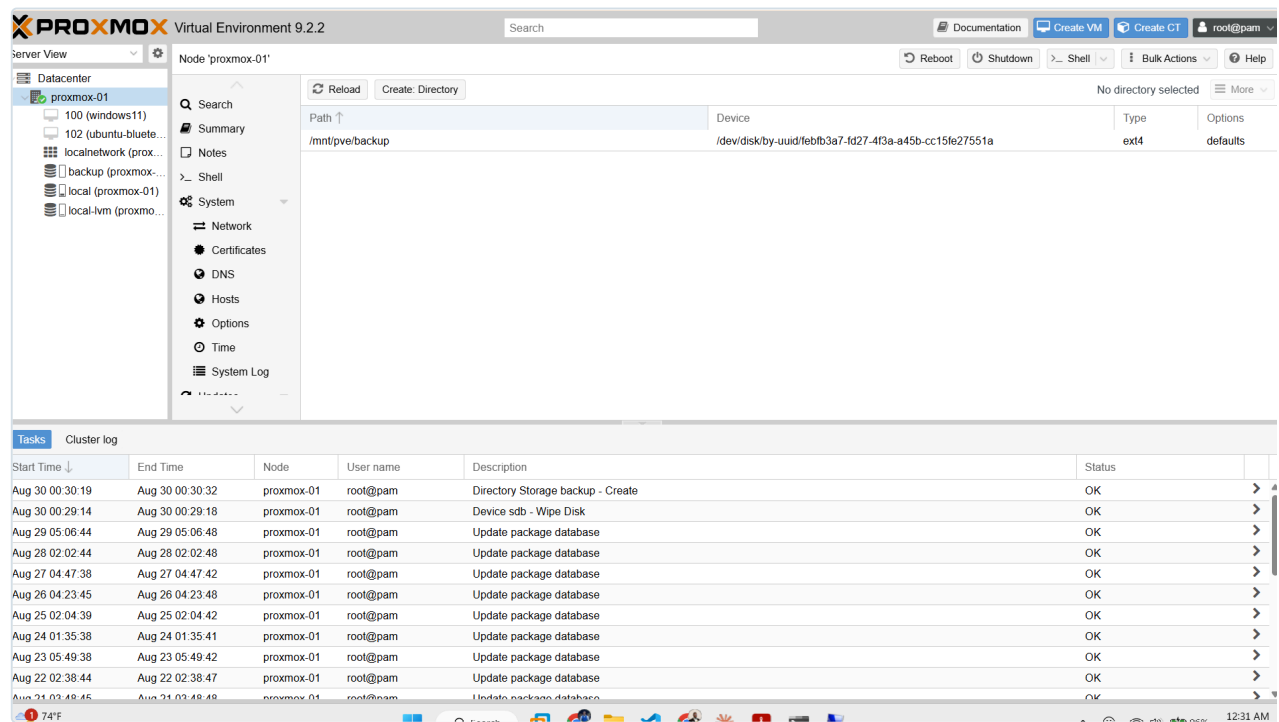


Figure 14.3: The `backup` Directory storage mounted at `/mnt/pve/backup` on the ext4 `sdb` disk. This is the dedicated backup target, separate from the disk that holds the running guests.

2. **Scheduled backup job:**

Setting	Value	Reason
Schedule	Daily 02:00	Off-hours, low activity
Selection	All guests	Auto-includes future VMs/CTs; no rule to forget
Mode	Snapshot	Backs up running guests with minimal disruption
Compression	ZSTD	Fast with good ratio
Retention	Keep last 3	Recent restore points without filling the disk
Target	backup (separate disk)	Survives primary-disk loss

3. **Verified** with an on-demand run: task completed **TASK OK** , backup files present under **backup** → Backups for both existing guests.

The screenshot shows the Proxmox VE 9.2.2 interface. In the left sidebar, the 'Datacenter' menu is expanded, and 'Backup' is selected. The main panel displays the 'Backup' configuration for the 'proxmox-01' node. The configuration is as follows:

Enabled	Node	Schedule	Next Run	Storage	Comment	Retention	Selection
☒	-- All --	02:00	2026-08-30 02:00:00	backup		keep-last=3	-- All --

Below the configuration, the 'Tasks' tab is active, showing a list of cluster log entries:

Start Time	End Time	Node	User name	Description	Status
ug 30 00:40:37		proxmox-01	root@pam	Backup Job	
ug 30 00:30:19	Aug 30 00:30:32	proxmox-01	root@pam	Directory Storage backup - Create	OK
ug 30 00:29:14	Aug 30 00:29:18	proxmox-01	root@pam	Device sdb - Wipe Disk	OK
ug 29 05:06:44	Aug 29 05:06:48	proxmox-01	root@pam	Update package database	OK
ug 28 02:02:44	Aug 28 02:02:48	proxmox-01	root@pam	Update package database	OK
ug 27 04:47:38	Aug 27 04:47:42	proxmox-01	root@pam	Update package database	OK

Figure 14.4: The backup job: enabled, all nodes, daily 02:00, target **backup** , retention keep-last=3, selection All. The running task at the bottom is the verification run.

What it protects, and what it does not

- **Protects:** all Proxmox VMs and containers (current and future). A lost or broken guest becomes a **restore**, not a rebuild.
- **Does not protect:** physical machines (e.g., the pfSense box, and a future physical DC02). Those need their own backup approach (for a Windows DC, the AD database is also protected by having a *second* DC, plus Windows Server Backup / system state).

Recovery

To restore a guest: Proxmox UI → **backup** storage → Backups → select the backup → Restore. Test restores periodically; a backup never verified by a restore is only a hope.

Future upgrade

Proxmox Backup Server (PBS) adds deduplication and incremental backups (far less space, faster). Not needed for a single node now; a Directory target is sufficient. Revisit if backup volume grows.

4. Capacity plan

RAM is the limit (~32 GiB). The rule that keeps the lab within budget:

Linux services run as LXC containers; Windows runs as full VMs.

LXC containers share the host kernel, a fraction of the RAM and disk of a VM. Windows cannot be a container, so it uses full VMs.

Guest	Type	Planned RAM
DC01 / DC02 (if virtual)	VM	2–4 GB
CA01	VM	2–4 GB
WKS01 / WKS02	VM	4 GB each
ANS01 (Ansible)	LXC	~1 GB
SIEM01 (Wazuh)	LXC/VM	~4 GB
MON01 (Grafana/Prometheus)	LXC	~2 GB
Kali	VM	~3 GB

All-running total is roughly 28 GB, within 32 GB, and guests can be powered on **on demand** rather than all at once. Verdict: the host comfortably carries the planned estate.

5. Machine naming convention

A consistent, role-based naming scheme (adopted 2026-08-30). Names state the role, so any machine is identifiable from its name alone. This supports asset inventory and audit (NIST CM-8).

Role	Name(s)
Domain controllers	DC01 , DC02
Certificate Authority	CA01
File / SQL servers	FS01 , SQL01
Workstations	WKS01 , WKS02
Wazuh / SIEM	SIEM01
Grafana / monitoring	MON01
Vault	VAULT01
Ansible controller	ANS01
Attack / offensive box	KALI01

Migration note: the first domain controller was originally named `SRV1` and was **renamed to `DC01`** on **2026-08-30** to match this convention. The rename was change-controlled: `dcdiag /test:dns` was clean before and after, the IP `192.168.50.2` and the domain `ad.biira.online` were unaffected, and the Netlogon service was restarted to re-register the DC's SRV records under the new name. The firewall alias retains its original name `SRV1_DC` (it points to the IP, not the hostname). Existing guests `100 (windows11)` and `102 (ubuntu-bluteam)` will be renamed to their role names when rebuilt/repurposed.

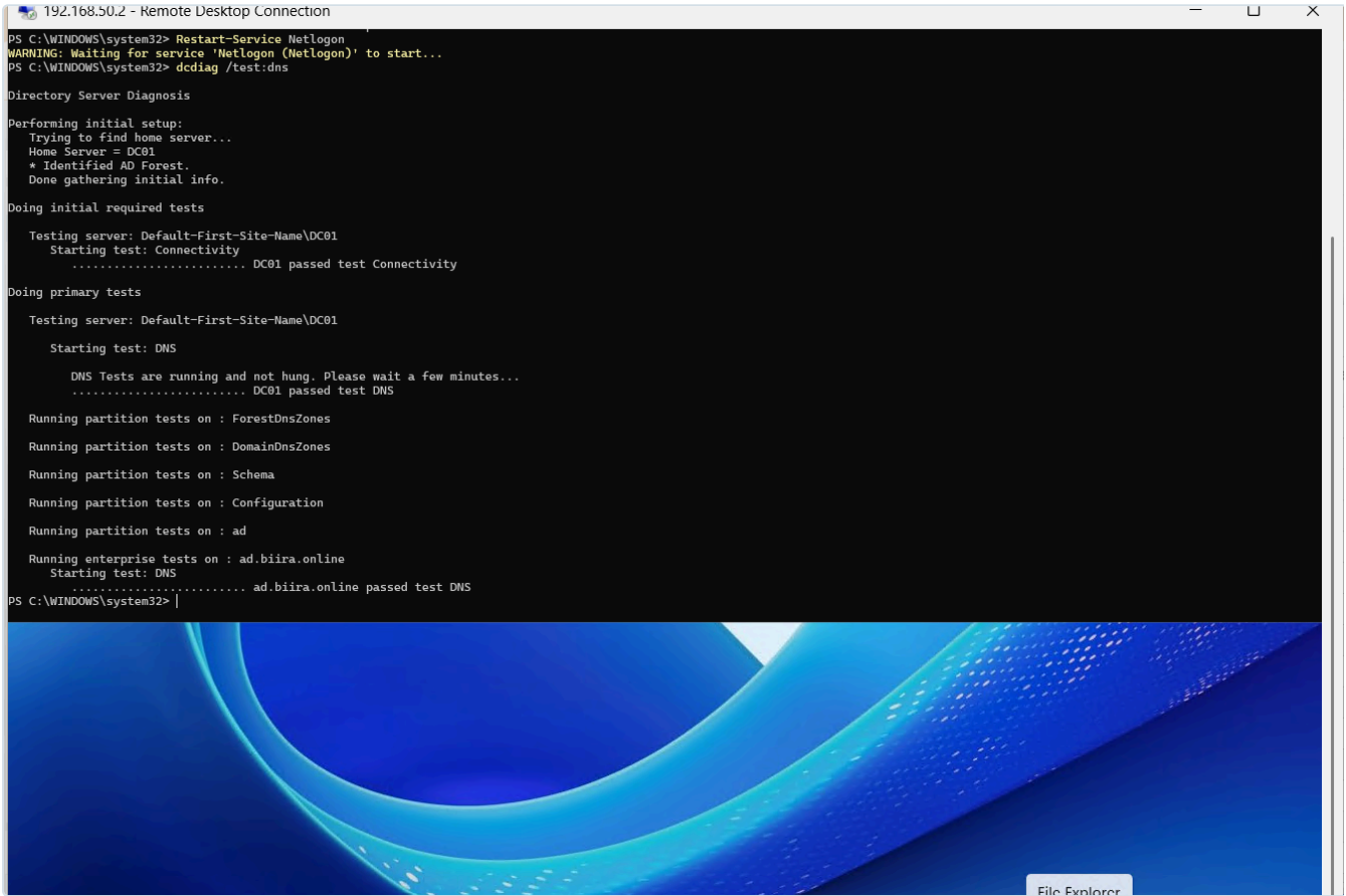


Figure 14.5: `dcdiag /test:dns` on `DC01` after the rename and Netlogon restart: Connectivity, DNS, all partition tests, and the enterprise `ad.biira.online` DNS test all pass. Evidence the rename left Active Directory healthy.

6. Standards alignment

What we did	Control
Independent, retained, scheduled backups	NIST SP 800-53 CP-9 (System Backup)
Documented restore procedure; periodic restore testing	NIST CP-10 (System Recovery) / CP-4 (Contingency testing)
Backups on separate disk from primary	Recoverability / defence against single-disk loss
Role-based naming for identifiable assets	NIST CM-8 (System Component Inventory)
This document itself	NIST CM-2/CM-6 (Baseline / documented configuration)

These are lab-appropriate implementations, not a claim of formal compliance.

